

WESTHAMPTON GABRESKI, NY, USA

WMO: 744865

Lat: 40.844N

Lon: 72.632W

Elev: 67

StdP: 14.66

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 14719

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.3	11.9	-5.7	4.1	12.3	-1.5	5.1	15.6	28.8	25.0	26.7	26.8	5.5	340	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	87.6	72.7	84.2	71.4	81.7	70.8	76.2	82.4	75.0	80.4	73.8	78.7	10.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.3	128.3	78.9	73.1	123.4	77.7	72.3	119.7	77.1	39.5	82.3	38.4	80.5	37.3	78.4	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.6	21.0	18.9	-1.7	93.6	5.5	3.7	-5.7	96.3	-8.9	98.4	-12.0	100.5	-16.0	103.2		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-2.4	78.6	5.4	1.6	-6.3	79.7	-9.5	80.6	-12.6	81.5	-16.5	82.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	29.8	31.2	37.3	46.7	56.1	65.8	72.2	71.2	64.5	53.5	44.0	35.6
	DBStd	16.58	10.00	8.41	8.07	6.93	6.69	6.16	5.25	5.29	6.55	8.00	8.25	9.07
	HDD50	2439	628	527	397	142	19	0	0	0	0	54	217	455
	HDD65	5806	1093	946	858	550	284	64	7	9	88	364	631	912
	CDD50	2721	1	1	5	43	210	474	687	657	436	164	36	7
	CDD65	611	0	0	0	1	10	88	229	200	74	9	0	0
	CDH74	4309	0	0	0	16	87	586	1736	1510	352	22	0	0
	CDH80	1006	0	0	0	3	13	143	472	330	43	2	0	0
Wind	WSAvg	8.5	9.8	9.6	10.1	9.3	8.0	7.5	7.1	6.6	7.2	8.4	8.8	9.2
Precipitation	PrecAvg	43.7	3.4	3.0	4.4	3.8	3.3	3.4	3.1	3.6	4.1	4.0	3.2	4.4
	PrecMax	57.9	6.7	5.4	12.3	7.0	6.7	8.4	5.7	9.3	8.0	18.7	5.7	8.0
	PrecMin	31.4	1.7	1.0	0.6	1.5	0.5	0.1	0.8	0.8	1.0	0.2	0.4	1.5
	PrecStd	6.3	1.3	1.2	2.6	1.6	1.5	2.3	1.3	2.1	1.8	3.5	1.4	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	55.5	64.3	75.4	81.6	89.7	93.4	90.2	84.4	76.9	65.9	61.2
		MCWB	52.9	46.9	53.2	57.5	65.9	72.5	74.9	73.9	72.0	66.9	58.8	56.1
	2%	DB	53.0	51.3	57.1	67.0	75.6	83.9	88.5	86.2	80.6	72.2	63.3	57.2
		MCWB	50.8	46.9	47.7	53.9	63.0	69.6	72.9	72.4	70.0	65.8	58.5	54.4
	5%	DB	49.4	48.3	53.5	62.6	71.9	80.5	84.6	83.5	77.8	69.9	61.2	54.6
		MCWB	46.1	44.4	47.4	51.8	61.6	68.2	71.7	71.8	68.9	64.0	56.9	52.1
	10%	DB	46.1	45.2	50.2	58.6	68.1	76.8	81.8	81.3	75.4	67.1	58.6	51.7
		MCWB	42.4	41.0	45.1	49.8	59.1	66.7	71.2	71.6	68.2	62.2	54.7	48.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	51.5	56.2	62.9	69.2	74.9	78.1	78.0	75.5	70.2	62.8	58.5
		MCDB	55.0	52.9	61.2	71.1	77.8	83.5	88.3	85.0	80.5	73.6	64.3	59.8
	2%	WB	50.9	48.1	51.3	57.4	66.4	72.8	76.1	76.4	73.7	67.8	60.3	55.3
		MCDB	52.6	50.3	54.1	62.5	73.1	80.2	83.4	81.9	77.2	70.7	62.4	56.7
	5%	WB	46.6	45.3	48.3	54.7	63.9	70.9	74.9	75.3	72.2	65.7	57.8	52.6
		MCDB	48.7	47.7	52.0	58.9	69.0	76.7	80.9	80.0	75.5	68.8	60.4	54.3
	10%	WB	43.1	41.3	45.9	52.5	61.6	69.2	73.7	74.1	70.4	63.4	55.3	48.5
		MCDB	45.7	44.7	49.3	56.5	66.4	74.2	78.9	78.4	73.9	66.2	58.1	51.1
Mean Daily Temperature Range	5% DB	MDBR	17.4	18.6	18.3	19.7	18.9	17.9	17.6	17.9	18.7	19.6	19.1	17.8
		MCDBR	19.6	21.2	22.3	26.0	24.1	22.6	21.1	19.7	19.3	20.2	18.9	18.1
		MCWBR	18.3	17.7	16.2	15.5	13.7	11.5	9.3	9.8	11.9	14.6	15.7	17.8
	5% WB	MCDBR	18.7	19.4	19.3	19.0	19.3	17.4	15.7	14.0	13.7	15.5	16.1	16.8
		MCWBR	18.4	18.2	16.2	13.4	11.9	9.9	8.1	8.1	9.7	13.0	15.2	17.7
Clear-Sky Solar Irradiance	taub	0.311	0.320	0.340	0.384	0.435	0.474	0.490	0.464	0.395	0.350	0.320	0.311	
	taud	2.509	2.484	2.470	2.367	2.259	2.193	2.155	2.225	2.386	2.519	2.555	2.552	
	Ebn at Noon	270	284	290	282	269	257	252	255	267	270	265	261	
	Edh at Noon	23	28	31	37	43	46	47	43	34	27	23	21	
All-Sky Solar Radiation	RadAvg	580	860	1193	1544	1728	1886	1906	1715	1397	953	656	493	
	RadStd	49	82	99	160	169	184	101	106	108	87	52	43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191	+106

Nomenclature: See separate page